1 Atmospheric Delay Modelling: Troposphere and Ionosphere

The undifferenced code and phase equations, the code observation and the phase observation of the Foundations document, each carry two propagation terms whose interiors are not states of the filter but *models*: the slant tropospheric delay T_r^i and the first-order slant ionospheric delay $\mu_s\, I_r^i$. The troposphere is split into a deterministic, a-priori *zenith hydrostatic* part that is computed and removed before the filter sees the measurement, plus a small *zenith wet* residual and a pair of horizontal gradients that ARE estimated (the state vector, the ZWD mapping). The ionosphere, in the uncombined parameterisation, is carried as one slant state per satellite (dynamic indexing); the models in this chapter supply its a-priori seed, an optional pseudo-observation that ties the relative slants to a Global Ionosphere Map, and the small second- and third-order corrections that the linear $1/f^2$ scaling cannot represent. The network producer additionally co-estimates its own spherical-harmonic VTEC field on a single thin shell (optionally two, Section 1.2.4), which lets it dispense with an external GIM entirely; that model is derived in Section 1.2.4, and the shared column layout it occupies in the network normal equations in the network mode chapter. This document is the derivation of those interiors: the mapping functions, the pressure reductions, the thin-shell geometry, the geomagnetic field, and the sign conventions that decide whether each term is added to or subtracted from the modelled range.

1.1 The slant tropospheric delay

The neutral atmosphere is non-dispersive at GNSS frequencies, so its delay is identical on code and phase and is absent from any ionosphere-free or geometry-free combination. The standard decomposition splits the slant delay into a hydrostatic ("dry") and a wet zenith delay, each carried to the line of sight by its own elevation mapping function, plus an azimuthal gradient [7]:

$$\boxed{T_r^i = m_{\mathrm{h}}(\varepsilon)\,\mathrm{ZHD} + m_{\mathrm{w}}(\varepsilon)\,\mathrm{ZWD} + m_{\mathrm{g}}(\varepsilon)\big(G_{\mathrm{N}}\cos\alpha + G_{\mathrm{E}}\sin\alpha\big).} \tag{1}$$

Of these, ZHD is supplied a priori and removed from the prefit residual; ZWD, G_{N} and G_{E} are estimated states whose Jacobian columns are exactly the mapping coefficients $m_{\mathrm{w}}(\varepsilon)$ and $m_{\mathrm{g}}(\varepsilon)\{\cos\alpha, \sin\alpha\}$ (`src/core/measurement/MeasurementModel.cpp`; the ZWD mapping). The remainder of this subsection derives each factor.

1.1.1 Saastamoinen zenith hydrostatic delay

The zenith hydrostatic delay is fixed to better than a millimetre by the total surface pressure alone, through the Saastamoinen reduction [6] as coded in `src/core/atmosphere/Troposphere.cpp`:

$$\boxed{\mathrm{ZHD} = \frac{0.0022768\ P}{1 - 0.00266\cos 2\phi - 0.00028\,h_{\mathrm{km}}},} \tag{2}$$

with P the total pressure in millibar at the antenna, ϕ the geodetic latitude, and $h_{\mathrm{km}} = h/1000$ the ellipsoidal height in kilometres (`Troposphere::computeSaastamoinenZHD`, denominator `f_lat`). The numerical coefficient $0.0022768\ \mathrm{m\,mbar^{-1}}$ folds the refractivity constant $k_1 = 77.604\ \mathrm{K\,mbar^{-1}}$ with the dry-gas constant and standard gravity; the denominator is the gravity-variation factor $f(\phi, h)$ that converts the column-integrated delay to its value at the local effective gravity.

When neither a surface met sensor nor the GPT3 grid (below) is available, P is taken from a standard-atmosphere barometric reduction of a configured sea-level pressure P_0 and temperature T_0 (`Troposphere.cpp`, `pressureFromHeightStdAtm`):

$$P(h) = P_0 \left(1 - \frac{L\,h}{T_0}\right)^{g/(RL)}, \qquad L = 0.0065\ \mathrm{K\,m^{-1}}, \quad \frac{g}{RL} \approx 5.25588, \tag{3}$$

with $g = 9.80665\ \mathrm{m\,s^{-2}}$ and $R = 287.05\ \mathrm{J\,kg^{-1}\,K^{-1}}$ the specific gas constant of dry air. The base $(1 - Lh/T_0)$ is clamped to $[0.1, 2.0]$ and the result to $[100, 1100]$ mbar as a numerical guard.

1.1.2 The Marini mapping form

Both the hydrostatic and wet mappings use the normalised three-parameter continued fraction of Marini, normalised to unity at zenith (`Troposphere::mariniMapping`):

$$\boxed{m(\varepsilon;\, a, b, c) = \cfrac{1 + \cfrac{a}{1 + \cfrac{b}{1 + c}}}{\sin\varepsilon + \cfrac{a}{\sin\varepsilon + \cfrac{b}{\sin\varepsilon + c}}}.} \tag{4}$$

The numerator is the denominator evaluated at $\varepsilon = 90^\circ$ ($\sin\varepsilon = 1$), so $m(90^\circ) = 1$ exactly; the only thing that distinguishes the hydrostatic from the wet mapping, and one model from another, is the triple (a, b, c). Two families are implemented.

Niell / GMF coefficients. The default mapping interpolates the Niell tables [5] of (a, b, c) given at five reference latitudes $15^\circ, 30^\circ, 45^\circ, 60^\circ, 75^\circ$, each as a mean plus an annually varying amplitude (the coefficient tables inside `Troposphere::getTroposphereMapping`). The seasonal value is

$$v(\phi, t) = \bar{v}(\phi) - \delta v(\phi)\,\cos\frac{2\pi\,(\mathrm{doy} - \mathrm{doy}_0)}{365.25}, \qquad \mathrm{doy}_0 = \begin{cases} 28 & \phi \geq 0 \\ 28 + 182.625 & \phi < 0 \end{cases} \tag{5}$$

(`Troposphere::getTroposphereMapping`, over the day of year returned by `Troposphere::getDayOfYear`). The *minus* sign and the southern-hemisphere half-year phase shift are both load-bearing: the amplitude term is subtracted, and the Southern Hemisphere is put in anti-phase by adding half a year to the phase reference rather than by flipping the sign. The latitude argument is the absolute latitude, linearly interpolated and clamped to $[15^\circ, 75^\circ]$; the wet coefficients carry mean values only (no annual term in this model).

A height correction lifts the hydrostatic mapping from sea level to the station, identical in the Niell and VMF3 paths (`Troposphere::getTroposphereMapping`):

$$m_\mathrm{h}(\varepsilon, h) = m_\mathrm{h}(\varepsilon, 0) + \left[\frac{1}{\sin\varepsilon} - m(\varepsilon;\, a_\mathrm{ht}, b_\mathrm{ht}, c_\mathrm{ht})\right] h_\mathrm{km}, \qquad \begin{aligned} a_\mathrm{ht} &= 2.53 \times 10^{-5}, \\ b_\mathrm{ht} &= 5.49 \times 10^{-3}, \\ c_\mathrm{ht} &= 1.14 \times 10^{-3}, \end{aligned} \tag{6}$$

with the same Marini form (4) evaluated at the height triple. Only the hydrostatic mapping receives this correction.

VMF3 coefficients. When configured for VMF3 [4], the wet/hydrostatic a-coefficients come from the GPT3 grid (next paragraph) and the b, c coefficients are synthesised on the fly by a degree- and order-12 spherical-harmonic expansion of their seasonal columns (`src/core/atmosphere/Vmf3.cpp`). A Cartesian-form associated Legendre recursion builds the real and imaginary solid harmonics V_{nm}, W_{nm} at the station co-latitude and longitude, the $91 = \sum_{n=0}^{12}(n+1)$ harmonic terms are accumulated against the stored `Vmf3Coeffs` tables, and each of $b_\mathrm{h}, c_\mathrm{h}, b_\mathrm{w}, c_\mathrm{w}$ is then evaluated as a five-coefficient mean-plus-annual-plus-semiannual harmonic — all three steps inside `Vmf3::computeMapping`:

$$v(\mathrm{doy}) = A_0 + A_1\cos\omega + B_1\sin\omega + A_2\cos 2\omega + B_2\sin 2\omega, \qquad \omega = \frac{2\pi\,\mathrm{doy}}{365.25}. \tag{7}$$

The synthesised (a, b, c) feed the very same Marini fraction (4), followed by the height correction (6). A failed grid query silently falls back to the Niell path so a single bad sample never aborts the run.

1.1.3 GPT3 empirical pressure, temperature and VMF3 a-coefficients

The GPT3 5° grid [4] supplies, per cell, the five harmonic coefficients of pressure, temperature, temperature lapse rate, geoid undulation, and the VMF3 hydrostatic and wet a-coefficients (`Gpt3::load`, in `atmosphere/Gpt3`). Each quantity is the same annual-plus-semiannual harmonic as (7) evaluated per cell (`Gpt3::harmonic`):

$$V(\mathrm{doy}) = a_0 + A_1\cos\omega + B_1\sin\omega + A_2\cos 2\omega + B_2\sin 2\omega. \tag{8}$$

The grid value sits at the cell's orthometric reference height H_s; it is carried to the station orthometric height $h_\mathrm{ortho} = h - N$ (undulation N) by a barometric pressure reduction and a linear temperature lapse (`Gpt3::query`):

$$P_\mathrm{stn} = P_\mathrm{grid}\big(1 - K_\mathrm{P}\,\Delta h\big)^{n_\mathrm{P}}, \quad T_\mathrm{stn} = T_\mathrm{grid} + \frac{\mathrm{d}T}{\mathrm{d}z}\,\Delta h, \qquad \begin{aligned} K_\mathrm{P} &= 2.26\times10^{-5}\,\mathrm{m}^{-1},\\ n_\mathrm{P} &= 5.225, \end{aligned} \tag{9}$$

with $\Delta h = h_\mathrm{ortho} - H_s$ and $\mathrm{d}T/\mathrm{d}z$ the grid lapse in $\mathrm{K\,km^{-1}}$ applied to Δh in kilometres. The exponent 5.225 is the same $g/(RL)$ ratio as (3), rounded for the grid. Four corners are read and combined by bilinear interpolation in latitude and (wrapped) longitude (also `Gpt3::query`); the resulting P_stn drives the Saastamoinen ZHD of (2), and the interpolated $a_\mathrm{h}, a_\mathrm{w}$ drive the VMF3 mapping.

1.1.4 The gradient mapping and a degenerate fallback

The azimuthal gradient in (1) uses the Chen–Herring mapping [1], coded identically in the Jacobian and the prefit residual (`MeasurementModel::buildJacobianRow`, `StateInitializer::computeRawResidual`):

$$\boxed{m_\mathrm{g}(\varepsilon) = \frac{1}{\sin\varepsilon\tan\varepsilon + 0.0032}, \qquad T^\mathrm{grad} = m_\mathrm{g}(\varepsilon)\big(G_\mathrm{N}\cos\alpha + G_\mathrm{E}\sin\alpha\big).} \tag{10}$$

The two partial derivatives $\partial\mathrm{T}/\partial G_\mathrm{N} = m_\mathrm{g}\cos\alpha$ and $\partial\mathrm{T}/\partial G_\mathrm{E} = m_\mathrm{g}\sin\alpha$ are exactly the Jacobian columns of the north/east gradient states (the state vector). Finally, a related cold-start guard applies to the *elevation* (hydrostatic/wet) mapping, not the gradient mapping above: when the receiver position is not yet usable (norm below 10^3 m, or non-finite) the geodetic latitude/height

needed by the Niell/VMF3 coefficients is unavailable, so the hydrostatic and wet mappings degrade to the geometry-only Black–Eisner form $1/(\sin\varepsilon + a/(\tan\varepsilon + b))$ (the fallback branch of `Troposphere::getTroposphereMapping`) purely so the engine never divides by an undefined elevation during cold start.

1.2 The slant ionospheric delay

The ionosphere is dispersive: to first order its excess phase advance and group delay are $\mp 40.3\,\mathrm{STEC}/f^2$, equal in magnitude and opposite in sign on phase versus code, where STEC is the slant total electron content in $\mathrm{el\,m^{-2}}$. The uncombined model carries this as a per-satellite slant state I_r^i referenced to GPS L1, and projects it onto every signal by the dispersion ratio.

1.2.1 The $1/f^2$ scaling and its sign

For a signal of frequency f on a state referenced to $f_{\mathrm{ref}} = f_{\mathrm{L1}}^{\mathrm{GPS}}$, the dispersion coefficient is

$$\boxed{\mu_s = \frac{f_{\mathrm{ref}}^2}{f_s^2}, \qquad \frac{\partial P_s}{\partial I} = +\mu_s, \qquad \frac{\partial \Phi_s}{\partial I} = -\mu_s,} \tag{11}$$

i.e. the slant-iono Jacobian column is $+\mu_s$ on code and $-\mu_s$ on phase (`MeasurementModel::buildJacobianRow`). This is the model interior of μ_s in the notation document; referencing every constellation's slant state to GPS L1 is what lets the between-satellite/between-receiver differencing operators (the differencing document) cancel the receiver iono datum cleanly. Because $f_{\mathrm{ref}} = f_{\mathrm{L1}}$, the L1 signal has $\mu = 1$ and the state *is* the L1 slant delay in metres.

1.2.2 IONEX / GIM thin-shell slant delay

A Global Ionosphere Map gives vertical TEC on a latitude/longitude grid at a single thin shell of height H above a base radius R_E. The slant delay along a ray at elevation ε follows from single-layer geometry (`computeIppGeometry` in `src/core/atmosphere/SphericalHarmonicIonosphere.cpp`, the shared helper that `GlobalIonosphereMap::slantIonoDelay` and the co-estimated spherical-harmonic VTEC model both call so the two never drift). The zenith angle z' at the ionospheric pierce point (IPP) obeys

$$\sin z' = \frac{R_E}{R_E + H}\cos\varepsilon, \qquad \text{mapping } M(\varepsilon) = \frac{1}{\cos z'}, \tag{12}$$

so that $\mathrm{STEC} = M(\varepsilon)\,\mathrm{VTEC}$ (`GlobalIonosphereMap::slantIonoDelay`). The earth-centred angle to the IPP is $\psi = \frac{\pi}{2} - \varepsilon - z'$, and the IPP latitude and longitude come from the spherical-triangle solution along the satellite azimuth α (`computeIppGeometry`). It propagates the receiver's geodetic latitude ϕ_r across the thin shell as though it were geocentric — an approximation worth well under a millimetre of L1 delay, and the frame in which the reconstructed IPP is later consumed by the higher-order model (30):

$$\phi_{\mathrm{IPP}} = \arcsin\!\big(\sin\phi_r\cos\psi + \cos\phi_r\sin\psi\cos\alpha\big), \tag{13}$$

$$\lambda_{\mathrm{IPP}} = \lambda_r + \arcsin\frac{\sin\psi\,\sin\alpha}{\cos\phi_{\mathrm{IPP}}}. \tag{14}$$

The VTEC at the IPP is interpolated bilinearly in space and, in time, *sun-fixed*: each bracketing map is sampled at the IPP longitude rotated by the Earth's rotation between the map

epoch and the query epoch, $\Delta\lambda = (t - t_{\mathrm{map}}) \cdot 360^\circ/86400$ s, before the two are linearly blended (`GlobalIonosphereMap::interpolateVtec`, over `GlobalIonosphereMap::sampleMap`). The slant delay in metres at frequency f then uses the dispersion constant

$$\boxed{I^{\mathrm{GIM}} = \mathrm{STEC} \cdot \frac{40.3 \times 10^{16}}{f^2} \quad [\mathrm{m}], \qquad \mathrm{STEC} = M(\varepsilon)\,\mathrm{VTEC}\ [\mathrm{TECU}],} \tag{15}$$

the factor 10^{16} converting VTEC from TECU to $\mathrm{el\,m^{-2}}$ and $40.3 = 40.3\ \mathrm{m\,Hz^2\,(el\,m^{-2})^{-1}}$ the first-order refractivity constant (both the constant and the metre conversion live in `GlobalIonosphereMap::slantIonoDelay`).

1.2.3 The GIM iono-weighting pseudo-observation

The GIM is too coarse to constrain the *absolute* slant of any one satellite, but it constrains the *relative* slants across a constellation well. This is imposed as a Kalman pseudo-observation (`Application::applyGimIonoUpdate`), one row per satellite that owns a slant state above the elevation cutoff. The innovation is the GIM slant minus the current state,

$$z_i = I_i^{\mathrm{GIM}} - I_r^i, \qquad \boldsymbol{H} = \boldsymbol{I}\ \text{(selection of the iono columns)}, \tag{16}$$

with the GIM evaluated at GPS L1 to match the state reference (`Application::applyGimIonoUpdate`). The structure is in the measurement-noise matrix. Per constellation g, with $\mathbf{1}$ the all-ones vector over that constellation's rows,

$$\boxed{\boldsymbol{R} = \bigoplus_{g \in \mathcal{G}} \Big(\mathrm{diag}(\sigma_i^2) + \sigma_c^2\, \mathbf{1}\mathbf{1}^{\mathrm{T}} \Big),} \tag{17}$$

(also `Application::applyGimIonoUpdate`). The rank-one $\sigma_c^2\,\mathbf{1}\mathbf{1}^{\mathrm{T}}$ block assigns an essentially unbounded variance to the per-constellation *common mode* of the innovations: that common mode is the un-modellable receiver iono datum (an arbitrary offset shared by all satellites of a constellation), so the pseudo-observation leaves it free and constrains only the between-satellite differences that the GIM actually knows. The update is applied through the same Joseph-form pseudo-update as every other measurement (the Kalman filter document); the iono process noise is documented there (the iono random-walk spectral density) and not repeated here.

Why frozen pivots are excluded. In the BSSD parameterisation each constellation's pivot satellite holds its slant-iono state frozen at value 0 with variance 0 as the rank-fixing datum. Such a row is skipped entirely (the frozen-pivot guard in `Application::applyGimIonoUpdate`). A zero-variance state receives zero Kalman gain and so could never benefit from the GIM; but it shares the common-mode $\mathbf{1}\mathbf{1}^{\mathrm{T}}$ block of (17) with its constellation peers. Were it included, its spurious innovation $I_{p_g}^{\mathrm{GIM}} - 0 \sim 5$–$30$ m would couple, through the off-diagonal of $\boldsymbol{S}^{-1}$, into the gain of every other satellite of that constellation, dragging the relative slants by the pivot's full absolute STEC. Excluding the pivot row is therefore not an optimisation but a correctness requirement: the common-mode block is meant to absorb the datum offset of the *remaining* rows, so the datum satellite must not itself appear as a measurement.

1.2.4 The co-estimated spherical-harmonic VTEC field

Rather than import an external map, the network producer can estimate the vertical TEC field itself — a truncated real spherical-harmonic expansion on the same kind of thin shell, solved jointly

with the satellite code biases. The coefficients are *shared* columns of the network normal equations, registered through `NetworkEstimator::registerVtecCoeff` and reduced onto the shared border like any other network-wide family (the network mode chapter); what follows is the model itself, its pseudo-observation, and its datum.

The switch is `network.estimate_iono_sh`. It defaults false, but it is part of the standing armed network posture and is set by most shipped network configurations, so it is the ionosphere model the producer actually runs on rather than an experiment. Arming it forces `network.estimate_sat_code_bias` on — the field is what makes the satellite code-bias slice observable — and suppresses external GIM loading, so the two ionosphere sources can never be applied to the same run (`config/GnssConfigLoadNetwork.cpp`; the SH-versus-GIM branch guards the `networkAppendGimIonoRows_` call site in `app/ApplicationNetworkAssembly`).

Basis and normalisation. With β the geographic latitude of the pierce point and s its sun-fixed longitude, the field is

$$\boxed{\mathrm{VTEC}(\beta, s) = \sum_{n=0}^{N} \sum_{m=0}^{n} \bar{P}_{nm}(\sin\beta)\left[a_{nm}\cos(ms) + b_{nm}\sin(ms)\right] \quad [\mathrm{TECU}],} \tag{18}$$

truncated triangularly at degree N (`estimation.global_vtec_sh.max_degree`), which gives $(N{+}1)^2$ real coefficients: one cosine term per $(n, m \geq 0)$ and one sine term per $(n, m \geq 1)$, enumerated once in a fixed order that the shared-column layout mirrors (`SphericalHarmonicIonosphere::coeffs`). The normalisation is

$$\bar{P}_{nm} = N_{nm}\, P_{nm}, \qquad N_{nm} = \sqrt{(2 - \delta_{0m})\,(2n+1)\,\frac{(n-m)!}{(n+m)!}}, \tag{19}$$

chosen so the real surface harmonics $\{\bar{P}_{nm}\cos(ms),\ \bar{P}_{nm}\sin(ms)\}$ are orthonormal under $(1/4\pi)\int(\cdot)\,\mathrm{d}\Omega$ [7]. Because (18) is linear in the coefficients, the basis values *are* the partial derivatives — $\partial\mathrm{VTEC}/\partial a_{nm}$ and $\partial\mathrm{VTEC}/\partial b_{nm}$ are exactly what `SphericalHarmonicIonosphere::basis` returns, so no numerical differentiation enters the Jacobian. One consequence is used repeatedly below: $\bar{P}_{00} \equiv 1$, so a_{00} *is* the global mean VTEC.

Frame and geometry. The expansion is carried in a sun-fixed frame, $s = \lambda_{\mathrm{IPP}} - \lambda_{\odot}$, with $\lambda_{\odot} = \alpha_{\odot} - \mathrm{GMST}$ the sub-solar geographic longitude (`meanSunLongitude`), so the field is quasi-static over a session instead of sweeping westward at the Earth's rotation rate; `estimation.global_vtec_sh.sun_fixed` selects geographic longitude instead. The pierce point and the thin-shell mapping $M(\varepsilon)$ that carries VTEC to STEC come from *the same* `computeIppGeometry` helper that `GlobalIonosphereMap::slantIonoDelay` calls, evaluated at `estimation.global_vtec_sh.layer_height_km` over base radius `earth_radius_km`. Equations (12), (13) and (14) are therefore the co-estimated model's geometry verbatim: the external map and the co-estimated field share one geometry implementation precisely so they cannot drift apart.

The pin row. The field observes nothing directly; it is tied to the estimated slant states by one pseudo-observation per line of sight (`Application::networkAppendShIonoRows_`). For every GPS or Galileo LOS above `ionosphere_weighting.elevation_cutoff_deg` whose slant state lives in

that station's local block, the row asserts that the slant state equals the field's prediction:

$$\boxed{\begin{aligned} & g_r^i \equiv I_r^i \;-\; \kappa\, M(\varepsilon) \sum_k \bar{Y}_k(\mathrm{IPP})\, c_k \;=\; 0, \qquad \kappa = \frac{40.3 \times 10^{16}}{f_{\mathrm{L1}}^2}, \\ & \frac{\partial g_r^i}{\partial I_r^i} = +1, \qquad \frac{\partial g_r^i}{\partial c_k} = -\,\kappa\, M(\varepsilon)\, \bar{Y}_k(\mathrm{IPP}). \end{aligned}} \tag{20}$$

Here c_k runs over the field coefficients, $\bar{Y}_k$ over the corresponding basis functions of (18), and κ is the TECU-to-L1-metre conversion already introduced in (15). The $+1$ lands in the station's private block and the $-\kappa M \bar{Y}_k$ terms in the *shared* border; that second write is the whole difference between an ionosphere prior and a joint estimate. The right-hand side is $-I_r^i$ at the current state, the shared coefficients entering the absolute solve at zero offset.

The measurement noise on these rows is *the same* common-mode-free per-constellation block as the GIM pseudo-observation, (17), built from the same `ionosphere_weighting.common_mode_sigma_m` over a per-LOS representativeness sigma

$$\sigma_i = \max\bigl(\texttt{floor_m},\ \sigma_{\mathrm{rep}}\, M(\varepsilon)\, \kappa\bigr), \qquad \sigma_{\mathrm{rep}} = \texttt{global_vtec_sh.representation_sigma_tecu}, \tag{21}$$

after which the block is whitened by the Cholesky factor of $\boldsymbol{R}$ into unit-information rows before it joins the normal equations. Reusing (17) is a modelling decision, not an implementation convenience. The rank-one $\sigma_c^2\, \mathbf{1}\mathbf{1}^{\mathrm{T}}$ block leaves the per-(receiver, constellation) iono datum free, so the receiver DCB rides in the common mode and the field constrains only the *between-satellite* slant structure — which is the only thing a global VTEC surface can be asked to know. Its corollary is that a_{00} is observable only weakly, through the spread of $M(\varepsilon)$ across the network, and that is what the datum below has to handle. GLONASS is excluded from the field entirely (`networkAppendShIonoRows_` skips it): its FDMA per-slot inter-frequency bias does not cancel in the common-mode block and would leak into the global coefficients.

The Kaula–Tikhonov prior as the estimability datum. Because the coefficients are shared columns, a coefficient whose pierce-point support has rotated over an ocean or polar gap is near-null in the reduced normal equations and would trip the all-or-nothing rank gate (`NetworkSbasis::assertNullSpaceRank`). A degree-dependent prior is therefore folded on *every* coefficient before the gate runs (`networkInjectShVtecPrior`):

$$\boxed{\begin{aligned} & \sigma_n = \frac{\texttt{kaula_scale_tecu}}{n} \quad (n \geq 1), \qquad \sigma_0 = \texttt{mean_vtec_prior_sigma_tecu}, \\ & \bar{c}_k = \begin{cases} \texttt{mean_vtec_prior_tecu}, & c_k = a_{00}, \\ 0, & \text{otherwise}, \end{cases} \end{aligned}} \tag{22}$$

combined in information space with an optional random-walk anchor toward the previous epoch's estimate, inflated by a per-degree spectral density $q_n = \texttt{q_per_degree}/n^2$ for $n \geq 1$. The two roles the prior plays must be kept apart. The Kaula term is *estimability*: it bounds every coefficient positive-definitely, so the declared null-space dimension stays equal to the number of constellations regardless of coverage, and it pulls unobserved coefficients back toward climatology each epoch instead of letting them random-walk away. The a_{00} term is a deliberately *loose* climatology anchor, and its looseness is the point: a_{00} trades against the absolute satellite-DCB level, whose only rigorous datum is the per-(constellation, signal) sum-to-zero constraint of the network mode chapter. Clamping a_{00} to an external mean VTEC would re-import the very GIM dependency the co-estimation exists to remove, and would bias every emitted satellite code bias by the difference.

Open extensions — default OFF, not yet field-validated. Two extensions of this model are built but off by default and still under validation; neither is the shipped behaviour.

(i) A second thin shell (`estimation.global_vtec_sh.layer2_max_degree`, default 0 = single layer) registers a second coefficient set at its own height, and the pin row (20) then sums the shells, each evaluated at *its own* pierce geometry, $\sum_l \kappa\, M_l(\varepsilon)\, \bar{Y}_k(\mathrm{IPP}_l)\, c_{k,l}$ (`ShLayer`, assembled by `shLayers`, which every consumer of the field loops so a shell can never be evaluated at another shell's geometry). The second shell's a_{00} anchor is deliberately *tight* — the mirror image of the argument above. The inter-layer split of the mean VTEC is quasi-null from ground-only geometry, since only the difference of the two shells' mapping profiles observes it, so the split must be held by prior information while the *total* mean stays data-determined through the first shell's loose anchor.

(ii) A per-(station, satellite) representativeness deviation (`estimation.global_vtec_sh.rep_error_tau_s`, default 0 = off) replaces the pin-to-zero of (20) with a pin to a tracked first-order Gauss–Markov deviation d, carried in vertical TECU so its level and steady state are elevation-stationary. Writing d^-, P^- for the aged prior, $\hat{u}, \hat{V}$ for the solved posterior mean and variance of the pin functional, and f^2 for the pin-row floor variance, the exact scalar posterior is

$$\hat{d} = d^- + \frac{P^-}{a}\big(\hat{u} - d^-\big), \qquad P^+ = P^-\,\frac{U + f^2}{U + a}, \qquad a = P^- + f^2, \quad U = \frac{\hat{V}\,a}{a - \hat{V}}, \tag{23}$$

with U the data-only information recovered from $\hat{V}$ (`shDevExactUpdate`). Two anti-ratchet treatments sit on top of it, also default OFF. `rep_error_demean` (`shDevDemean`) removes the information-weighted global and per-station means of the carried priors once per epoch, because a deviation component that is common in vertical TECU has row signature $\kappa\, M(\varepsilon)$ — *identical* to the a_{00} column's — so the pair is exactly quasi-null and the harvest–age–re-assert cycle ratchets it; the de-mean enforces the state's own contract, that it carries per-LOS structure only. `rep_error_d5_keep` (`shDevD5Retap`) splits the slip response by provenance: a raw-detector-confirmed slip still *erases* the deviation, keeping faith with the rule that a stiff time-linked carry must have a wired slip response (the cycle-slip chapter), while an innovation-gate-only re-tap keeps the level and merely re-taps the variance to the Gauss–Markov steady state. The warm-LOS counts that arm the de-mean are constants in the seam, not configuration; the loader warns that both treatments are inert unless `rep_error_tau_s` is positive, and that arming the erase-policy split without the de-mean re-opens the ratchet.

1.2.5 Klobuchar broadcast seed

For a single-frequency a-priori only — the cold-start seed of the slant state, never a dual-frequency correction — the broadcast Klobuchar half-cosine model is available [3] (`Ionosphere::computeKlobucharDelay`). With elevation and azimuth in semicircles, the earth-centred angle, geomagnetic IPP latitude, local time and obliquity are

$$\psi = \frac{0.0137}{\varepsilon/180 + 0.11} - 0.022, \qquad \phi_i = \mathrm{clip}_{\pm 0.416}\big(\phi/180 + \psi\cos\alpha\big), \tag{24}$$

$$\phi_m = \phi_i + 0.064\cos\big((\lambda_i - 1.617)\pi\big), \qquad F = 1 + 16\,(0.53 - \varepsilon/180)^3, \tag{25}$$

with $\lambda_i = \lambda/180 + \psi\sin\alpha/\cos(\phi_i\pi)$ and local time $t_i = 43200\,\lambda_i + t \pmod{86400}$. Cubic polynomials in the geomagnetic latitude ϕ_m give the amplitude and period,

$$\mathrm{AMP} = \sum_{k=0}^{3} a_k\,\phi_m^k\ (\geq 0), \qquad \mathrm{PER} = \sum_{k=0}^{3} b_k\,\phi_m^k\ (\geq 72000\ \mathrm{s}), \tag{26}$$

and the zenith delay is the truncated half-cosine, mapped to slant by F and scaled to the requested frequency by the f_{L1}^2/f^2 dispersion ratio of (11) (the tail of `Ionosphere::computeKlobucharDelay`):

$$I^{\mathrm{Klob}} = c\,F\Big(5\times10^{-9} + \mathrm{AMP}\left[1 - \tfrac{X^2}{2} + \tfrac{X^4}{24}\right]\Big)\frac{f_{\mathrm{L1}}^2}{f^2}, \quad X = \frac{2\pi(t_i - 50400)}{\mathrm{PER}},\ |X| < 1.57, \tag{27}$$

and the constant 5×10^{-9} s night term alone when $|X| \geq 1.57$. The leading c converts the modelled signal delay from seconds to metres.

1.3 Higher-order ionospheric corrections

The $1/f^2$ scaling captures the first-order term; the residual second- and third-order delays scale as $1/f^3$ and $1/f^4$ and are applied as deterministic corrections to the modelled range (`HigherOrderIonosphere::slantHigherOrderComponents`). The second order is the magnetic-field–weighted delay

$$\boxed{I_2^P = 7527\,c\,\mathrm{STEC}\,\frac{\boldsymbol{B}\cdot\hat{\boldsymbol{k}}}{f^3}, \qquad I_2^\Phi = -\tfrac{1}{2}\,I_2^P,} \tag{28}$$

where STEC is in $\mathrm{el\,m^{-2}}$, $\boldsymbol{B}$ is the geomagnetic flux density in tesla at the IPP, $\hat{\boldsymbol{k}}$ the receiver→satellite unit vector, and c the speed of light (the constant $7527\,c = e\,A/(2\pi m_e) \approx 2.2564 \times 10^{12}$ folds the electron charge, the 40.3-family refractivity constant A, and the electron mass; the constants and the $-\frac{1}{2}$ phase factor are both in `HigherOrderIonosphere::slantHigherOrderComponents`). The third order depends on the path integral of N_e^2, which a shape factor η relates to the F2-peak density and STEC (also `HigherOrderIonosphere::slantHigherOrderComponents`):

$$\boxed{I_3^P = 2437\,\eta\,N_{\max}\,\frac{\mathrm{STEC}}{f^4}, \qquad N_{\max} = \kappa\,\mathrm{STEC}, \qquad I_3^\Phi = -\tfrac{1}{3}\,I_3^P,} \tag{29}$$

with $\eta = 0.66$ the Hoque–Jakowski shape factor [2], $2437 = 3A^2/8$, and $N_{\max} = \kappa\,\mathrm{STEC}$ the linear Hoque–Jakowski fit ($\kappa = (20-6)\times10^{12}/((4.55-1.38)\times10^{18})$ $\mathrm{el\,m^{-3}/(el\,m^{-2})}$) of the F2-peak electron density to slant STEC. The third-order term is thus *linear* in $N_{\max}\,\mathrm{STEC}$, not in STEC^2. Both corrections are sub-millimetre at L1 but enter with the correct sign on each measurement type and survive between-satellite differencing because they are satellite-specific.

1.3.1 The geomagnetic dipole field

The field at the IPP is a centred, tilted magnetic dipole evaluated in ECEF (`MagneticField::fieldAtEcef`):

$$\boxed{\boldsymbol{B}(\boldsymbol{r}) = B_0\left(\frac{R_E}{r}\right)^3\left(\hat{\boldsymbol{m}} - 3(\hat{\boldsymbol{m}}\cdot\hat{\boldsymbol{r}})\,\hat{\boldsymbol{r}}\right),} \tag{30}$$

with $r = \|\boldsymbol{r}\|$, $\hat{\boldsymbol{r}} = \boldsymbol{r}/r$, $B_0 = 30\times10^{-6}$ T the equatorial surface field, $R_E = 6371.2$ km the IGRF reference radius, and $\hat{\boldsymbol{m}}$ the unit axis pointing to the geographic location of the geomagnetic north pole, IGRF-13 epoch 2020.0 at $(80.65°, -72.68°)$ (`MagneticField::dipoleAxisEcef`). With $\hat{\boldsymbol{m}}$ directed at the Arctic geomagnetic pole (which is the dipole's *south* pole), the sign convention in (30) makes $\boldsymbol{B}$ point horizontally toward magnetic north at the magnetic equator, matching compass behaviour; this sign is what fixes the sign of $\boldsymbol{B}\cdot\hat{\boldsymbol{k}}$ and hence of I_2 in (28). The IPP ECEF position

itself is reconstructed as a spherical-to-Cartesian map of the geocentric IPP latitude/longitude at the IONEX base-plus-shell radius, since IONEX is a thin-shell model (`HigherOrderIonosphere::slantHigherOrderComponents`).

Together with the deterministic ZHD removal, the estimated ZWD/gradient states, and the uncombined slant-iono state, these models populate every atmospheric term of the undifferenced observation equation; the differencing operators that subsequently eliminate the receiver-common pieces of them are the subject of the differencing document.

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